

IELTS

Workbook



Reading

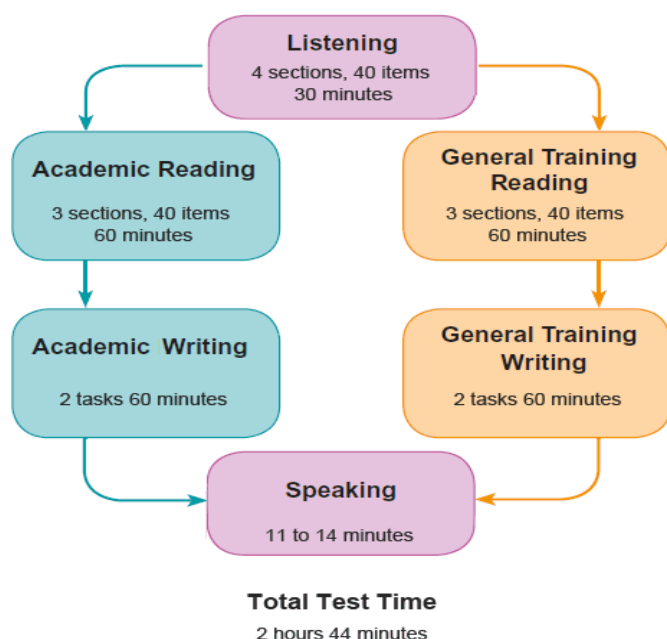
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Preface

IELTS (International English Language Testing System) is a worldwide recognized official English exam. The IELTS is administered by the British Council, IDP: IELTS Australia and Cambridge English Language Assessment. Its aim is to evaluate the English level of the people that take it. This is one of the most accepted English exams by higher education institutions to certify English language proficiency.



IELTS Reading: Academic Module – 3 long texts of an academic nature. General Training Module - Mixture of long and short texts of a more general nature, as well as texts related to work situation.

Reading Paper is Assessed on:

- Identifying the overall purpose
- Follow key arguments in a text
- Identify opinions and attitudes
- Locate specific information
- Distinguish main ideas from details.
- Extract information from a text
- complete a diagram, summary, table

Reading Types of Questions:

- Matching Heading Questions
- True/ False/Not Given or Yes / No/ Not Given.
- Matching Paragraph Information Question.
- Summary Completion Question
- Sentence Completion Question
- Multiple Choice Question
- List Selection
- Choosing a Title
- Categorization Question
- Matching Sentence Endings
- **Table Completion**
- Flow chart Completion Question
- Short Answer Question

Table Completion questions are quite like flow-chart Questions but tend to cover a larger part of the reading passage. Sometimes it may cover the entire passage. Here the headings given to the paragraphs in your notes will help you locate your answers. Such questions demand for specific answers so; the passage needs to be thoroughly read. These questions are commonly encountered for categorisation-based passages like theories proposed by different scientists, different types of processes. Also, the information may not be given in the same order as in the passage. With our tips & practise papers we are sure you are going to enjoy solving such question and get correct answers

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TO SOLVE TABLE COMPLETION QUESTIONS

Table Completion questions are like flow-chart Questions but tend to cover a larger part of the reading passage. Sometimes it may cover the entire passage. Here the headings given to the paragraphs in your notes will help you locate your answers. Such questions demand for specific answers so; the passage needs to be thoroughly read. These questions are commonly encountered for categorisation-based passages like theories proposed by different scientists, different types of processes. Also, the information may not be given in the same order as in the passage. With our tips & practise papers we are sure you are going to enjoy solving such question and get correct answers

GUIDELINES TO SOLVE:

1. Skim read the entire passage to get the overall idea of the passage. (5 min)
2. Understand what kind of information is given in each paragraph. You can give headings to each paragraph for your self-understanding. This way it's easy for you to locate the paragraph which contains the information asked.
3. Study the table completion task. Pay attention to what kind of details are required.
4. Look for the paragraphs which have this text and for keywords and paraphrases from the question.
5. Usually the words required to fill in the blanks are specific details like numbers, dates, definitions, objects.
6. Fill in the gaps with suitable words considering the word limit mentioned in the question. Carefully examine the Table, before and after the words you have found.
7. Spend less than 5 min to solve table completion questions.

**Let's look at the application of the above-mentioned guidelines
for the following passage below.**

Solved Reading Passage

SHEET GLASS MANUFACTURE: THE FLOAT PROCESS

Glass, which has been made since the time of the Mesopotamians and Egyptians, is little more than a mixture of sand, soda ash and lime. When heated to about 1500 degrees Celsius (°C) this becomes a molten mass that hardens when slowly cooled. The first successful method for making clear, flat glass involved spinning. This method was very effective as the glass had not touched any surfaces between being soft and becoming hard, so it stayed perfectly unblemished, with a 'fire finish'. However, the process took a long time and was labour intensive.

Nevertheless, demand for flat glass was very high and glassmakers across the world were looking for a method of making it continuously. The first continuous ribbon process involved squeezing molten glass through two hot rollers, similar to an old mangle. This allowed glass of virtually any thickness to be made non-stop, but the rollers would leave both sides of the glass marked, and these would then need to be ground and polished. This part of the process rubbed away around 20 per cent of the glass, and the machines were very expensive.

The float process for making flat glass was invented by Alistair Pilkington. This process allows the manufacture of clear, tinted and coated glass for buildings, and clear and tinted glass for vehicles. Pilkington had been experimenting with improving the melting process, and in 1952 he had the idea of using a bed of molten metal to form the flat glass, eliminating altogether the need for rollers within the float bath. The metal had to melt at a temperature less than the hardening point of glass (about 600°C), but could not boil at a temperature below the temperature of the molten glass (about 1500°C). The best metal for the job was tin.

The rest of the concept relied on gravity, which guaranteed that the surface of the molten metal was perfectly flat and horizontal. Consequently, when pouring molten glass onto the molten tin, the underside of the glass would also be perfectly flat. If the glass were kept hot enough, it would flow over the molten tin until the top surface was also flat, horizontal and perfectly parallel to the bottom surface. Once the glass cooled to 604°C or less it was too hard to mark and could be transported out of the cooling zone by rollers. The glass settled to a thickness of six millimetres because of surface tension interactions between the glass and the tin. By fortunate coincidence, 60 per cent of the flat glass market at that time was for six-millimetre glass.

Pilkington built a pilot plant in 1953 and by 1955 he had convinced his company to build a full-scale plant. However, it took 14 months of non-stop production, costing the company £100,000 a month, before the plant produced any usable glass. Furthermore, once they succeeded in making marketable flat glass, the machine was turned off for a service to prepare it for years of continuous production. When it started up again it took another four months to get the process right again. They finally succeeded in 1959 and there are now float plants all over the world, with each able to produce around 1000 tons of glass every day, non-stop for around 15 years.

Float plants today make glass of near optical quality. Several processes - melting, refining, homogenising - take place simultaneously in the 2000 tonnes of molten glass in the furnace. They occur in separate zones in a complex glass flow driven by high temperatures. It

adds up to a continuous melting process, lasting as long as 50 hours, that delivers glass smoothly and continuously to the float bath, and from there to a coating zone and finally a heat treatment zone, where stresses formed during cooling are relieved.

The principle of float glass is unchanged since the 1950s. However, the product has changed dramatically, from a single thickness of 6.8 mm to a range from sub-millimetre to 25 mm, from a ribbon frequently marred by inclusions and bubbles to almost optical perfection. To ensure the highest quality, inspection takes place at every stage. Occasionally, a bubble is not removed during refining, a sand grain refuses to melt, a tremor in the tin puts ripples into the glass ribbon. Automated on-line inspection does two things. Firstly, it reveals process faults upstream that can be corrected. Inspection technology allows more than 100 million measurements a second to be made across the ribbon, locating flaws the unaided eye would be unable to see. Secondly, it enables computers downstream to steer cutters around flaws.

Float glass is sold by the square meter, and at the final stage computers translate customer requirements into patterns of cuts designed to minimize waste.

Complete the table below. Choose **NO MORE THAN TWO WORDS** from the passage for each answer. **The TWO questions below are solved for you.**

Early methods of producing flat glass

Method	Advantages	Disadvantages
1. ____ (1) ____	Glass remained ____ (2) ____	- Slow ____ (3) ____
2. Ribbon	- Could produce glass sheets of varying ____ (4) ____ - Non-stop process	- Glass was ____ (5) ____ 20% of glass rubbed away - Machines were expensive

Solved Examples:

- The above table specifically focuses on the early methods of producing flat glass. These methods are mentioned in the first 2 paragraphs of the passage. The first blank falls under methods category. There were only 2 such methods, spinning and ribbon. The second option says ribbon method so the first one should be **spinning process**. This indicates that the word for the fill in the gap is **spinning**.

2. The second blank falls under Advantages of the spinning process category. The answer can be traced from the following text belonging to the first paragraph which says, ‘This method was very effective as the glass had not touched any surfaces between being soft and becoming hard, so it **stayed perfectly unblemished**, with a ‘fire finish’. In the table the synonym used for stayed is the word remained. This indicates that the word for the fill in the gap is **unblemished**.

Example has been done for you above, now try to complete the table below. Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

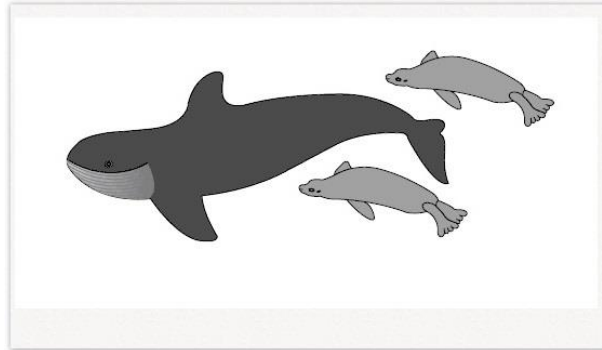
Early methods of producing flat glass

Method	Advantages	Disadvantages
3. ____ (1) ____	Glass remained ____ (2) ____	- Slow ____ (3) ____
4. Ribbon	- Could produce glass sheets of varying ____ (4) ____ - Non-stop process	- Glass was ____ (5) ____ 20% of glass rubbed away - Machines were expensive

Practise Reading Passage 1.1

WHAT DO WHALES FEEL?

An examination of the functioning of the senses in cetaceans, the group of mammals comprising whales, dolphins and porpoises.



Some of the senses that we and other terrestrial mammals take for granted are either reduced or absent in cetaceans or fail to function well in water. For example, it appears from their brain structure that toothed species are unable to smell. Baleen species, on the other hand, appear to have some related brain structures but it is not known whether these are functional. It has been speculated that, as the blowholes evolved and migrated to the top of the head, the neural pathways serving sense of smell may have been nearly all sacrificed. Similarly, although at least some cetaceans have taste buds, the nerves serving these have degenerated or are rudimentary.

The sense of touch has sometimes been described as weak too, but this view is probably mistaken. Trainers of captive dolphins and small whales often remark on their animals' responsiveness to being touched or rubbed, and both captive and free ranging cetacean individuals of all species (particularly adults and calves, or members of the same subgroup) appear to make frequent contact. This contact may help to maintain order within a group, and stroking or touching are part of the courtship ritual in most species. The area around the blowhole is also particularly sensitive and captive animals often object strongly to being touched there.

The sense of vision is developed to different degrees in different species. Baleen species studied at close quarters underwater – specifically a grey whale calf in captivity for a year, and free-ranging right whales and humpback whales studied and filmed off Argentina and Hawaii – have obviously tracked objects with vision underwater, and they can apparently see moderately well both in water and in air. However, the position of the eyes so restricts the field of vision in baleen whales that they probably do not have stereoscopic vision.

On the other hand, the position of the eyes in most dolphins and porpoises suggests that they have stereoscopic vision forward and downward. Eye position in freshwater dolphins, which often swim on their side or upside down while feeding, suggests that what vision they have is stereoscopic forward and upward. By comparison, the bottlenose dolphin has an extremely keen

vision in water. Judging from the way it watches and tracks airborne flying fish, it can apparently see fairly well through the air–water interface as well. And although preliminary experimental evidence suggests that their in-air vision is poor, the accuracy with which dolphins leap high to take small fish out of a trainer’s hand provides anecdotal evidence to the contrary.

Such variation can no doubt be explained with reference to the habitats in which individual species have developed. For example, vision is obviously more useful to species inhabiting clear open waters than to those living in turbid rivers and flooded plains. The South American bottlenose and Chinese beiji, for instance, appear to have very limited vision, and the Indian susu are blind, their eyes reduced to slits that probably allow them to sense only the direction and intensity of light.

Although the senses of taste and smell appear to have deteriorated, and vision in water appears to be uncertain, such weaknesses are more than compensated for by cetaceans’ well-developed acoustic sense. Most species are highly vocal, although they vary in the range of sounds, they produce, and many forage for food using echolocation. Large baleen whales primarily use the lower frequencies and are often limited in their repertoire. Notable exceptions are the nearly song-like choruses of bowhead whales in summer and the complex, haunting utterances of the humpback whales. Toothed species, in general, employ more of the frequency spectrum, and produce a wider variety of sounds than baleen species (though the sperm whale apparently produces a monotonous series of high-energy clicks and little else). Some of the more complicated sounds are clearly communicative, although what role they may play in the social life and ‘culture’ of cetaceans has been more the subject of wild speculation than of solid science.

1. echolocation: the perception of objects by means of sound wave echoes.

Questions 1-7

Complete the table below. Choose NO MORE THAN THREE WORDS from Reading Passage.

SENSE	SPECIES	ABILITY	COMMENTS
Smell	toothed	no	evidence from brain structure
	baleen	not certain	related brain structures are present
Taste	some types	poor	nerves linked to their(1)..... are underdeveloped
Touch	all	yes	region around the blowhole very sensitive
Vision	(2).....	Yes	probably do not have stereoscopic vision
	Dolphins, porpoises	Yes	probably have stereoscopic vision(3)..... and(3).....
	(4).....	Yes	probably have stereoscopic vision forward and upward
	Bottlenose dolphin	Yes	exceptional in(5)..... and good in air–water interface
	boutu and beiji	Poor	have limited vision
	Indian susu	No	probably only sense direction and intensity of light
Hearing	most large baleen	Yes	usually use(6).....; repertoire limited
	(7)..... whales and(7)..... whales	Yes	song-like
	toothed	Yes	use more of frequency spectrum; have wider repertoire

Practise Reading Passage 1.2

MICRO-ENTERPRISE CREDIT FOR STREET YOUTH

"I am from a large, poor family and for many years we have done without breakfast. Ever since I joined the Street Kids International program, I have been able to buy my family sugar and buns for breakfast. I have also bought myself decent second-hand clothes and shoes."

Doreen Soko

'We've had business experience. Now I am confident to expand what we've been doing. I've learnt cash management, and the way of keeping money so we save for re-: investment. Now business is a part of our lives. As well, we didn't know each other before - now we've made new friends.'

Fan Kaoma

Participants in the Youth Skills Enterprise Initiative Program, Zambia

Introduction

Although small-scale business training and credit programs have become more common throughout the world, relatively little attention has been paid to the need to direct such opportunities to young people. Even less attention has been paid to children living on the street or in difficult circumstances.

Over the past nine years, Street Kids International (S.K.I.) has been working with partner organizations in Africa, Latin America and India to support the economic lives of street children. The purpose of this paper is to share some of the lessons S.K.I. and our partners have learned.

Background

Typically, children do not end up on the streets due to a single cause, but to a combination of factors: a dearth of adequately funded schools, the demand for income at home, family breakdown and violence. The street may be attractive to children as a place to find adventurous play and money. However, It is also a place where some children are exposed, with little or no protection, to exploitative employment, urban crime, and abuse.

Children who work on the streets are generally involved in unskilled, labour-intensive tasks which require long hours, such as shining shoes, carrying goods, guarding, or washing cars, and informal trading. Some may also earn income through begging, or through theft and other illegal activities. At the same time, there are street children who take pride in supporting themselves and their families and who often enjoy their work. Many children may choose entrepreneurship because it allows them a degree of independence, is less exploitative than many forms of paid employment, and is flexible enough to allow them to participate in other activities such as education and domestic tasks.

Street Business Partnerships

S.K.I. has worked with partner organizations in Latin America, Africa and India to develop innovative opportunities for street children to earn income.

- The S.K.I. Bicycle Courier Service first started in the Sudan. Participants in this enterprise were supplied with bicycles, which they used to deliver parcels and messages, and which they were required to pay for gradually from their wages. A similar program was taken up in Bangalore, India.
- Another successful project, The Shoeshine Collective, was a partnership program with the Y.W.C.A. in the Dominican Republic. In this project, participants were lent money to purchase shoeshine boxes. They were also given a safe place to store their equipment, and facilities for individual savings plans.
- The Youth Skills Enterprise Initiative in Zambia is a joint program with the Red Cross Society and the Y.W.C.A. Street youths are supported to start their own small business through business training, life skills training and access to credit.

Lessons learned

The following lessons have emerged from the programs that S.K.I. and partner organizations have created.

- Being an entrepreneur is not for everyone, nor for every street child. Ideally, potential participants will have been involved in the organisation's programs for at least six months, and trust and relationship building will have already been established.
- The involvement of the participants has been essential to the development of relevant programs. When children have had a major role in determining procedures, they are more likely to abide by and enforce them.
- It is critical for all loans to be linked to training programs that include the development of basic business and life skills.
- There are tremendous advantages to involving parents or guardians in the program, where such relationships exist. Home visits allow staff the opportunity to know where the participants live, and to understand more about each individual's situation.
- Small loans are provided initially for purchasing fixed assets such as bicycles, shoe shine kits and basic building materials for a market stall. As the entrepreneurs gain experience, the enterprises can be gradually expanded and consideration can be given to increasing loan amounts. The loan amounts in S.K.I. programs have generally ranged from US\$30-\$100.

- All S.K.I. programs have charged interest on the loans, primarily to get the entrepreneurs used to the concept of paying interest on borrowed money. Generally the rates have been modest (lower than bank rates).

Conclusion

There is a need to recognize the importance of access to credit for impoverished young people seeking to fulfil economic needs. The provision of small loans to support the entrepreneurial dreams and ambitions of youth can be an effective means to help them change their lives. However, we believe that credit must be extended in association with other types of support that help participants develop critical life skills as well as productive businesses.

Questions 1-4

Complete the table below. Choose **NO MORE THAN THREE WORDS** from the reading passage.

Country	Organisation Involved	Type of Project	Support provided
____(1)____ and ____(1)____	• S.K.I	Courier service	• Provision of ____(2)____
Dominican	• S.K.I • Y.W.C.A.	____(3)____	• Loans • Storage facilities • Savings plans
Zambia	• S.K.I • The Red Cross • Y.W.C.A.	Setting up small businesses	• Business training • ____ (4) ____ training • Access to credit

Practise Reading Passage 1.3

LACK OF SLEEP

Section A

It is estimated that the average man or woman needs between seven-and-a-half and eight hours' sleep a night. Some can manage on a lot less. Baroness Thatcher, for example, was reported to be able to get by on four hours' sleep a night when she was Prime Minister of Britain. Dr Jill Wilkinson, senior lecturer in psychology at Surrey University and co-author of 'Psychology in Counselling and Therapeutic Practice', states that healthy individuals sleeping less than five hours or even as little as two hours in every 24 hours are rare, but represent a sizeable minority.

Section B

The latest beliefs are that the main purposes of sleep are to enable the body to rest and replenish, allowing time for repairs to take place and for tissue to be regenerated. One supporting piece of evidence for this rest-and-repair theory is that production of the growth hormone somatotropin, which helps tissue to regenerate, peaks while we are asleep. Lack of sleep, however, can compromise the immune system, muddle thinking, cause depression, promote anxiety and encourage irritability.

Section C

Researchers in San Diego deprived a group of men of sleep between 1am and 5am on just one night and found that levels of their bodies' natural defences against viral infections had fallen significantly when measured the following morning. 'Sleep is essential for our physical and emotional well-being and there are few aspects of daily living that are not disrupted by the lack of it', says Professor William Regelson of Virginia University, a specialist in insomnia. 'Because it can seriously undermine the functioning of the immune system, sufferers are vulnerable to infection.'

Section D

For many people, lack of sleep is rarely a matter of choice. Some have problems getting to sleep, others with staying asleep until the morning. Despite popular belief that sleep is one long event, research shows that, in an average night, there are five stages of sleep and four cycles, during which the sequence of stages is repeated. In the first light phase, the heart rate and blood pressure go down and the muscles relax. In the next two stages, sleep gets progressively deeper. In stage four, usually reached after an hour, the slumber is so deep that, if awoken, the sleeper would be confused and disorientated. It is in this phase that sleep-walking can occur, with an average episode lasting no more than 15 minutes. In the fifth stage, the rapid eye movement (REM) stage, the heartbeat quickly gets back to normal levels, brain activity accelerates to daytime heights and above and the eyes move constantly beneath closed lids as if the sleeper is looking at something. During this stage, the body is almost paralysed. This REM phase is also the time when we dream.

Section E

Sleeping patterns change with age, which is why many people over 60 develop insomnia. In America, that age group consumes almost half the sleep medication on the market. One theory for the age-related change is that it is due to hormonal changes. The temperature rise occurs at daybreak in the young, but at three or four in the morning in the elderly. Age aside, it is estimated that roughly one in three people suffer some kind of sleep disturbance. Causes can be anything from pregnancy and stress to alcohol and heart disease. Smoking is a known handicap to sleep, with one survey showing that ex-smokers got to sleep in 18 minutes rather than their earlier average of 52 minutes.

Section F

Apart from self-help therapy such as regular exercise, there are psychological treatments, including relaxation training and therapy aimed at getting rid of pre-sleep worries and anxieties. There is also sleep reduction therapy, where the aim is to improve sleep quality by strictly regulating the time people go to bed and when they get up. Medication is regarded by many as a last resort and often takes the form of sleeping pills, normally benzodiazepines, which are minor tranquillisers.

Section G

Professor Regelson advocates the use of melatonin for treating sleep disorders. Melatonin is a naturally secreted hormone, located in the pineal gland deep inside the brain. The main function of the hormone is to control the body's biological clock, so we know when to sleep and when to wake. The gland detects light reaching it through the eye; when there is no light, it secretes the melatonin into the bloodstream, lowering the body temperature and helping to induce sleep. Melatonin pills contain a synthetic version of the hormone and are commonly used for jet lag as well as for sleep disturbance. John Nicholls, sales manager of one of America's largest health food shops, claims that sales of the pill have increased dramatically. He explains that it is sold in capsules, tablets, lozenges and mixed with herbs. It is not effective for all insomniacs, but many users have weaned themselves off sleeping tablets as a result of its application.

Questions 1-4

Complete the table below. Choose NO MORE THAN TWO WORDS from the passage for each answer.

TREATMENTS FOR SLEEPING DISORDERS.	
Therapy	Function
1. _____(1)_____	
Example: regular exercise	
2. Psychological treatments	
Example A: _____(2)_____	To get rid of sleep pre-worries.
Example B: sleep reduction therapy	To improve _____ (3) _____ by regulating their sleeping and wake up timings.
3. Medication	
Example: _____(4)_____	Tranquillizers

Practise Reading Passage 1.4

A REMARKABLE BEETLE

Some of the most remarkable beetles are the dung beetles, which spend almost their whole lives eating and breeding in dung’.

More than 4,000 species of these remarkable creatures have evolved and adapted to the world’s different climates and the dung of its many animals. Australia’s native dung beetles are scrub and woodland dwellers, specializing in coarse marsupial droppings and avoiding the soft cattle dung in which bush flies and buffalo flies breed.

In the early 1960s George Bornemissza, then a scientist at the Australian Government’s premier research organisation, the Commonwealth Scientific and Industrial Research Organisation (CSIRO), suggested that dung beetles should be introduced to Australia to control dung-breeding flies. Between 1968 and 1982, the CSIRO imported insects from about 50 different species of dung beetle, from Asia, Europe and Africa, aiming to match them to different climatic zones in Australia. Of the 26 species that are known to have become successfully integrated into the local environment, only one, an African species released in northern Australia, has reached its natural boundary.

Introducing dung beetles into a pasture is a simple process: approximately 1,500 beetles are released; a handful at a time, into fresh cow pats in the cow pasture. The beetles immediately disappear beneath the pats digging and tunneling and, if they successfully adapt to their new environment, soon become a permanent, self-sustaining part of the local ecology. In time they multiply and within three or four years the benefits to the pasture are obvious.

Dung beetles work from the inside of the pat so they are sheltered from predators such as birds and foxes. Most species burrow into the soil and bury dung in tunnels directly underneath the pats, which are hollowed out from within. Some large species originating from France excavate tunnels to a depth of approximately 30 cm below the dung pat. These beetles make sausage-shaped brood chambers along the tunnels. The shallowest tunnels belong to a much smaller Spanish species that buries dung in chambers that hang like fruit from the branches of a pear tree. South African beetles dig narrow tunnels of approximately 20 cm below the surface of the pat. Some surface-dwelling beetles, including a South African species, cut perfectly-shaped balls from the pat, which are rolled away and attached to the bases of plants.

For maximum dung burial in spring, summer and autumn, farmers require a variety of species with overlapping periods of activity. In the cooler environments of the state of Victoria, the large French species (2.5 cms long) is matched with smaller (half this size), temperate-climate Spanish species. The former are slow to recover from the winter cold and produce only one or two generations of offspring from late spring until autumn. The latter, which multiply rapidly in early spring, produce two to five generations annually. The South African ball-rolling species, being a

subtropical beetle, prefers the climate of northern and coastal New South Wales where it commonly works with the South African tunneling species. In warmer climates, many species are active for longer periods of the year.

Dung beetles were initially introduced in the late 1960s with a view to controlling buffalo flies by removing the dung within a day or two and so preventing flies from breeding. However, other benefits have become evident. Once the beetle larvae have finished pupation, the residue is a first-rate source of fertiliser. The tunnels abandoned by the beetles provide excellent aeration and water channels for root systems. In addition, when the new generation of beetles has left the nest the abandoned burrows are an attractive habitat for soil-enriching earthworms. The digested dung in these burrows is an excellent food supply for the earthworms, which decompose it further to provide essential soil nutrients. If it were not for the dung beetle, chemical fertiliser and dung would be washed by rain into streams and rivers before it could be absorbed into the hard earth, polluting water courses and causing blooms of blue-green algae. Without the beetles to dispose of the dung, cow pats would litter pastures making grass inedible to cattle and depriving the soil of sunlight. Australia's 30 million cattle each produce 10-12 cow pats a day. This amounts to 1.7 billion tones a year, enough to smother about 110,000 sq km of pasture, half the area of Victoria.

Dung beetles have become an integral part of the successful management of dairy farms in Australia over the past few decades. A number of species are available from the CSIRO or through a small number of private breeders, most of whom were entomologists with the CSIRO's dung beetle unit who have taken their specialized knowledge of the insect and opened small businesses in direct competition with their former employer.

Glossary

- 1. *dung:- the droppings or excreta of animals***
- 2. *cow pats:- droppings of cows***

Questions 1-5

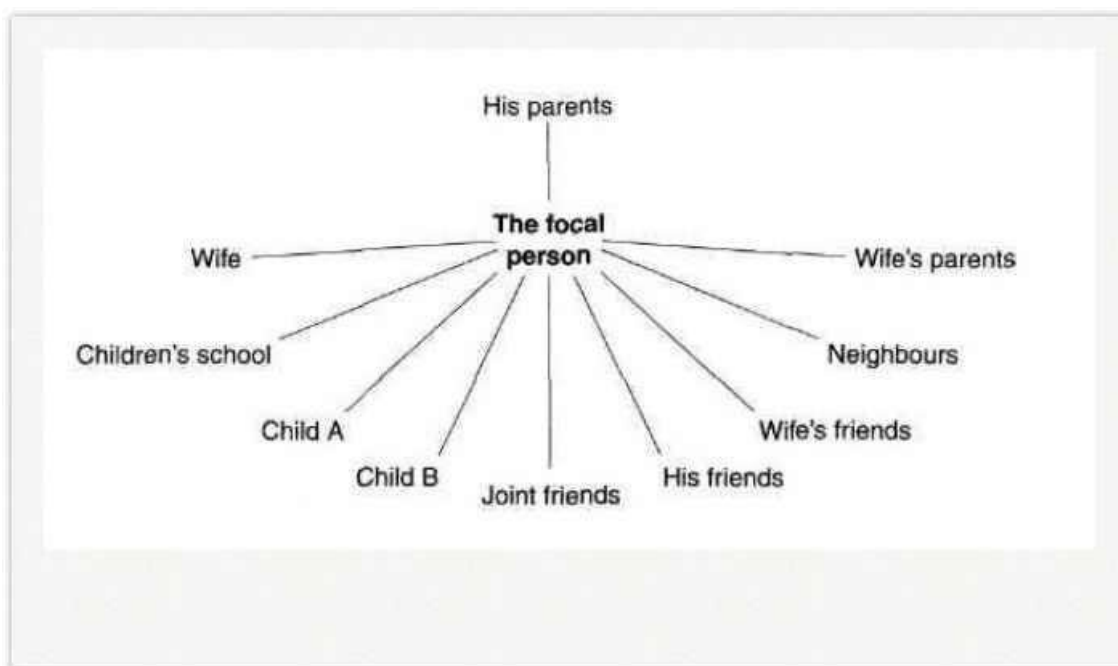
Complete the table below. Choose **NO MORE THAN THREE WORDS OR A NUMBER** from the above reading passage for each answer.

Species	Size	Preferred climate	Complimentary species	Start of active period	Number of generations per year
French	2.5 cm	Cool	Spanish	Late spring	1-2
Spanish	1.25 cm	1_____		2_____	3_____
South African ball roller		4_____	5_____		

Practise Reading Passage 1.5

THE CONCEPT OF ROLE THEORY

Any individual in any situation occupies a role in relation to other people. The particular individual with whom one is concerned in the analysis of any situation is usually given the name of *focal person*. He has the *focal role* and can be regarded as sitting in the middle of a group of people, with whom he interacts in some way in that situation. This group of people is called his *role set*. For instance, in the family situation, an individual's role set might be shown as in Figure 6.



The role set should include all those with whom the individual has more than trivial interactions.

Role definition

The definition of any individual's role in any situation will be a combination of the role expectations that the members of the role set have of the focal role. These expectations are often occupationally denned, sometimes even legally so. The role definitions of lawyers and doctors are fairly clearly defined both in legal and in cultural terms. The role definitions of, say, a film star or bank manager, are also fairly clearly defined in cultural terms, too clearly perhaps. Individuals often find it hard to escape from the role that cultural traditions have defined for them. Not only with doctors or lawyers is the required role behavior so constrained that if you are in that role for long it eventually becomes part of you, part of your personality. Hence, there is some likelihood that all accountants will be alike or that all blondes are similar – they are forced that way by the expectations of their role.

It is often important that you make it clear what your particular role is at a given time. The means of doing this are called, rather obviously, role signs. The simplest of role signs is a uniform. The number of stripes on your arm or pips on your shoulder is a very precise role definition which allows you to do certain very prescribed things in certain situations. Imagine yourself questioning a stranger on a dark street at midnight without wearing the role signs of a policeman!

In social circumstances, dress has often been used as a role sign to indicate the nature and degree of formality of any gathering and occasionally the social status of people present. The current trend towards blurring these role signs in dress is probably democratic, but it also makes some people very insecure. Without role signs, who is to know who has what role?

Place is another role sign. Managers often behave very differently outside the office and in it, even to the same person. They use a change of location to indicate a change in role from, say, boss to friend. Indeed, if you wish to change your roles you must find some outward sign that you are doing so or you won't be permitted to change – the subordinate will continue to hear you as his boss no matter how hard you try to be his friend. In very significant cases of role change, e.g. from a soldier in the ranks to officer, from bachelor to married man, the change of role has to have a very obvious sign, hence rituals. It is interesting to observe, for instance, some decline in the emphasis given to marriage rituals. This could be taken as an indication that there is no longer such a big change in role from single to married person, and therefore no need for a public change in sign.

In organizations, office signs and furniture are often used as role signs. These and other perquisites of status are often frowned upon, but they may serve a purpose as a kind of uniform in a democratic society; roles without signs often lead to confused or differing expectations of the role of the focal person.

Role ambiguity

Role ambiguity results when there is some uncertainty in the minds, either of the focal person or of the members of his role set, as to precisely what his role is at any given time. One of the crucial expectations that shape the role definition is that of the individual, the focal person himself. If his occupation of the role is unclear, or if it differs from that of the others in the role set, there will be a degree of role ambiguity. Is this bad? Not necessarily, for the ability to shape one's own role is one of the freedoms that many people desire, but the ambiguity may lead to role stress which will be discussed later on. The virtue of job descriptions is that they lessen this role ambiguity.

Unfortunately, job descriptions are seldom complete role definitions, except at the lower end of the scale. At middle and higher management levels, they are often a list of formal jobs and duties that say little about the more subtle and informal expectations of the role. The result is, therefore, to give the individual an uncomfortable feeling that there are things left unsaid, i.e. to heighten the sense of role ambiguity.

Looking at role ambiguity from the other side, from the point of view of the members of the role set, lack of clarity in the role of the focal person can cause insecurity, lack of confidence, irritation and even anger among members of his role set. One list of the roles of a manager identified the following: executive, planner, policy maker, expert, controller of rewards and punishments, counselor, friend, teacher. If it is not clear, through role signs of one sort or another, which role is currently the operational one, the other party may not react in the appropriate way — we may, in fact, hear quite another message if the focal person speaks to us, for example, as a teacher and we hear her as an executive.

Questions 13-16

Complete the following table with **NO MORE THAN TWO WORDS**.

Role sign	Function	Example of the situation
Dress	Indicates ____ (1) ____ and degree of Formality	Social circumstances
Place	Determines the ____ (2) ____ in case of role change	Marriage traditions
Furniture and office signs	Regarded as a ____ (3) ____ in the democratic society	____ (4) ____

Practice Reading Passage 1.6

THE DISCOVERY OF URANUS

Someone once put forward an attractive though unlikely theory. Throughout the Earth's annual revolution around the sun, there is one point of space always hidden from our eyes. This point is the opposite part of the Earth's orbit, which is always hidden by the sun. Could there be another planet there, essentially similar to our own, but always invisible?

If a space probe today sent back evidence that such a world existed it would cause not much more sensation than Sir William Herschel's discovery of a new planet, Uranus, in 1781. Herschel was an extraordinary man — no other astronomer has ever covered so vast a field of work — and his career deserves study. He was born in Hanover in Germany in 1738, left the German army in 1757, and arrived in England the same year with no money but quite exceptional music ability. He played the violin and oboe and at one time was organist in the Octagon Chapel in the city of Bath. Herschel's was an active mind, and deep inside he was conscious that music was not his destiny; he, therefore, read widely in science and the arts, but not until 1772 did he come across a book on astronomy. He was then 34, middle-aged by the standards of the time, but without hesitation he embarked on his new career, financing it by his professional work as a musician. He spent years mastering the art of telescope construction, and even by present-day standards, his instruments are comparable with the best.

Serious observation began in 1774. He set himself the astonishing task of 'reviewing the heavens', in other words, pointing his telescope to every accessible part of the sky and recording what he saw. The first review was made in 1775; the second, and most momentous, in 1780-81. It was during the latter part of this that he discovered Uranus. Afterwards, supported by the royal grant in recognition of his work, he was able to devote himself entirely to astronomy. His final achievements spread from the sun and moon to remote galaxies (of which he discovered hundreds), and papers flooded from his pen until his death in 1822. Among these, there was one sent to the Royal Society in 1781, entitled *An Account of a Comet*. In his own words:

On Tuesday the 13th of March, between ten and eleven in the evening, while I was examining the small stars in the neighbourhood of H Geminorum, I perceived one that appeared visibly larger than the rest; being struck with its uncommon magnitude, I compared it to H Geminorum and the small star in the quartile between Auriga and Gemini, and finding it to be much larger than either of them, suspected it to be a comet.

Herschel's care was the hallmark of a great observer; he was not prepared to jump any conclusions. Also, to be fair, the discovery of a new planet was the last thought in anybody's mind. But further observation by other astronomers besides Herschel revealed two curious facts. For the comet, it showed a remarkably sharp disc; furthermore, it was moving so slowly that it was thought to be a great distance from the sun, and comets are only normally visible in the immediate vicinity of the sun. As its orbit came to be worked out the truth dawned that it was a

new planet far beyond Saturn's realm, and that the 'reviewer of the heavens' had stumbled across an unprecedented prize. Herschel wanted to call it Georgian sidus (Star of George) in honour of his royal patron King George III of Great Britain. The planet was later for a time called Herschel in honour of its discoverer. The name Uranus, which was first proposed by the German astronomer Johann Elert Bode, was in use by the late 19th century.

Uranus is a giant in construction, but not so much in size; its diameter compares unfavourably with that of Jupiter and Saturn, though on the terrestrial scale it is still colossal. Uranus' atmosphere consists largely of hydrogen and helium, with a trace of methane. Through a telescope, the planet appears as a small bluish-green disc with a faint green periphery. In 1777, while recording the occultation of a star behind the planet, the American astronomer James L. Elliot discovered the presence of five rings encircling the equator of Uranus. Four more rings were discovered in January 1986 during the exploratory flight of Voyager 2. In addition to its rings, Uranus has 15 satellites ('moons'), the last 10 discovered by Voyager 2 on the same flight; all revolve about its equator and move with the planet in an east-west direction. The two largest moons, Titania and Oberon, were discovered by Herschel in 1787. The next two, Umbriel and Ariel, were found in 1851 by the British astronomer William Lassell. Miranda, thought before 1986 to be the innermost moon, was discovered in 1948 by the American astronomer Gerard Peter Kuiper.

Glossary:

'Occultation': in astronomy, when one object passes in front of another and hides the second from view, especially, for example, when the moon comes between an observer and a star or planet.

'Voyager 2': an unmanned spacecraft sent on a voyage past Saturn, Uranus and Jupiter in 1986; during which it sent back information about these planets to scientists on earth.

Questions 1-6

Complete the table below. Write a date for each answer.

Event	Date
William Herschel was born	(1) _____
Herschel began investigating astronomy	(2) _____
Discovery of the planet Uranus	(3) _____
Discovery of the moons Titania and Oberon	(4) _____
First discovery of Uranus' rings	(5) _____
Discovery of the last 10 moons of Uranus	(6) _____

Practise Reading Passage 1.7

ENVIRONMENTAL MANAGEMENT

Section A

The role of governments in environmental management is difficult but inescapable. Sometimes, the state tries to manage the resources it owns, and does so badly. Often, however, governments act in an even more harmful way. They actually subsidise the exploitation and consumption of natural resources. A whole range of policies, from farm- price support to protection for coal-mining, do environmental damage and (often) make no economic sense. Scrapping them offers a two-fold bonus: a cleaner environment and a more efficient economy. Growth and environmentalism can actually go hand in hand, if politicians have the courage to confront the vested interest that subsidies create.

Section B

No activity affects more of the earth's surface than farming. It shapes a third of the planet's land area, not counting Antarctica, and the proportion is rising. World food output per head has risen by 4 per cent between the 1970s and 1980s mainly as a result of increases in yields from land already in cultivation, but also because more land has been brought under the plough. Higher yields have been achieved by increased irrigation, better crop breeding, and a doubling in the use of pesticides and chemical fertilisers in the 1970s and 1980s.

Section C

All these activities may have damaging environmental impacts. For example, land clearing for agriculture is the largest single cause of deforestation; chemical fertilisers and pesticides may contaminate water supplies; more intensive farming and the abandonment of fallow periods tend to exacerbate soil erosion; and the spread of mono-culture and use of high-yielding varieties of crops have been accompanied by the disappearance of old varieties of food plants which might have provided some insurance against pests or diseases in future. Soil erosion threatens the productivity of land in both rich and poor countries. The United States, where the most careful measurements have been done, discovered in 1982 that about one-fifth of its farmland is losing topsoil at a rate likely to diminish the soil's productivity. The country subsequently embarked upon a program to convert 11 per cent of its cropped land to meadow or forest. Topsoil in India and China is vanishing much faster than in America.

Section D

Government policies have frequently compounded the environmental damage that farming can cause. In the rich countries, subsidies for growing crops and price supports for farm output drive up the price of land. The annual value of these subsidies is immense: about \$250 billion, or more than all World Bank lending in the 1980s. To increase the output of crops per acre, a farmer's easiest option is to use more of the most readily available inputs: fertilisers and pesticides. Fertiliser use doubled in Denmark in the period 1960-1985 and increased in The Netherlands by 150 per cent. The quantity of pesticides applied has risen too; by 69 per cent in 1975-1984 in Denmark, for example, with a rise of 115 per cent in the frequency of application in the three years from 1981.

In the late 1980s and early 1990s some efforts were made to reduce farm subsidies. The most dramatic example was that of New Zealand, which scrapped most farm support in 1984. A study of the environmental effects, conducted in 1993, found that the end of fertiliser subsidies had been followed by a fall in fertiliser use (a fall compounded by the decline in world commodity prices, which cut farm incomes). The removal of subsidies also stopped land-clearing and overstocking, which in the past had been the principal causes of erosion. Farms began to diversify. The one kind of subsidy whose removal appeared to have been bad for the environment was the subsidy to manage soil erosion,

In less enlightened countries, and in the European Union, the trend has been to reduce rather than eliminate subsidies, and to introduce new payments to encourage farmers to treat their land in environmentally friendlier ways, or to leave it fallow. It may sound strange but such payments need to be higher than the existing incentives for farmers to grow food crops. Farmers, however, dislike being paid to do nothing. In several countries they have become interested in the possibility of using fuel produced from crop residues either as a replacement for petrol (as ethanol) or as fuel for power stations (as biomass). Such fuels produce far less carbon dioxide than coal or oil, and absorb carbon dioxide as they grow. They are therefore less likely to contribute to the greenhouse effect. But they are rarely competitive with fossil fuels unless subsidised - and growing them does no less environmental harm than other crops.

Section E

In poor countries, governments aggravate other sorts of damage. Subsidies for pesticides and artificial fertilisers encourage farmers to use greater quantities than are needed to get the highest economic crop yield. A study by the International Rice Research Institute Of pesticide use by farmers in South East Asia found that, with pest-resistant varieties of rice, even moderate applications of pesticide frequently cost farmers more than they saved. Such waste puts farmers on a chemical treadmill: bugs and weeds become resistant to poisons, so next year's poisons must be more lethal. One cost is to human health, Every year some 10,000 people die from pesticide poisoning, almost all of them in the developing countries, and another 400,000 become seriously ill. As for artificial fertilisers, their use world-wide increased by 40 per cent per unit of farmed land between the mid 1970s and late 1980s, mostly in the developing countries. Overuse of fertilisers may cause farmers to stop rotating crops or leaving their land fallow. That, In turn, may make soil erosion worse.

Section F

A result of the Uruguay Round of world trade negotiations is likely to be a reduction of 36 per cent in the average levels of farm subsidies paid by the rich countries in 1986-1990. Some of the world's food production will move from Western Europe to regions where subsidies are lower or non-existent, such as the former communist countries and parts of the developing world. Some environmentalists worry about this outcome. It will undoubtedly mean more pressure to convert natural habitat into farmland. But it will also have many desirable environmental effects. The intensity of farming in the rich world should decline, and the use of chemical inputs will diminish. Crops are more likely to be grown in the environments to which they are naturally suited. And more farmers in poor countries will have the money and the incentive to manage

their land in ways that are sustainable in the long run. That is important. To feed an increasingly hungry world, farmers need every incentive to use their soil and water effectively and efficiently.

Questions 1-4

Complete the table below using the information in sections B and C of Reading Passage.

Choose your answers A-G from the box below the table and write them.

Agricultural Practise	Environmental damage that may result
_____ (1) _____	Deforestation
_____ (2) _____	Degraded water supply
More intensive farming	_____ (3) _____
Expansion of monoculture	_____ (4) _____

- A. Abandonment of fallow period
- B. Disappearance of old plant varieties
- C. Increased use of chemical inputs
- D. Increased irrigation
- E. Insurance against pests and diseases
- F. Soil erosion
- G. Clearing land for cultivation

Practise Reading Passage 1.8

OBTAINING LINGUISTIC DATA

- A. Many procedures are available for obtaining data about a language. They range from a carefully planned, intensive field investigation in a foreign country to a casual introspection about one's mother tongue carried out in an armchair at home.
- B. In all cases, someone has to act as a source of language data - an informant. Informants are (ideally) native speakers of a language, who provide utterances for analysis and other kinds of information about the language (e.g. translations, comments about correctness, or judgements on usage). Often, when studying their mother tongue, linguists act as their own informants, judging the ambiguity, acceptability, or other properties of utterances against their own intuitions. The convenience of this approach makes it widely used, and it is considered the norm in the generative approach to linguistics. But a linguist's personal judgements are often uncertain, or disagree with the judgements of other linguists, at which point recourse is needed to more objective methods of enquiry, using non-linguists as informants.

The latter procedure is unavoidable when working on foreign languages, or child speech.

- C. Many factors must be considered when selecting informants - whether one is working with single speakers (a common situation when languages have not been described before), two people interacting, small groups or large-scale samples. Age, sex, social background and other aspects of identity are important, as these factors are known to influence the kind of language used. The topic of conversation and the characteristics of the social setting (e.g. the level of formality) are also highly relevant, as are the personal qualities of the informants (e.g. their fluency and consistency). For larger studies, scrupulous attention has been paid to the sampling theory employed, and in all cases, decisions have to be made about the best investigative techniques to use.
- D. Today, researchers often tape-record informants. This enables the linguist's claims about the language to be checked and provides a way of making those claims more accurate ('difficult' pieces of speech can be listened to repeatedly). But obtaining naturalistic, good quality data is never easy. People talk abnormally when they know they are being recorded, and sound quality can be poor. A variety of tape-recording procedures have thus been devised to minimise the 'observer's paradox' (how to observe the way people behave when they are not being observed). Some recordings are made without the speakers being aware of the fact - a procedure that obtains very natural data, though ethical objections must be anticipated. Alternatively, attempts can be made to make the speaker forget about the recording, such as keeping the tape recorder out of sight, or using radio microphones. A useful technique is to introduce a topic that quickly involves the speaker, and stimulates a natural language style (e.g. asking older informants about how times have changed in their locality).

- E. An audio tape recording does not solve all the linguist's problems, however. Speech is often unclear and ambiguous. Where possible, therefore, the recording has to be supplemented by the observer's written comments on the non-verbal behaviour of the participants, and about the context in general. A facial expression, for example, can dramatically alter the meaning of what is said. Video recordings avoid these problems to a large extent, but even they have limitations (the camera cannot be everywhere), and transcriptions always benefit from any additional commentary provided by an observer.
- F. Linguists also make great use of structured sessions, in which they systematically ask their informants for utterances that describe certain actions, objects or behaviours. With a bilingual informant, or through use of an interpreter, it is possible to use translation techniques ('How do you say table in your language?'). A large number of points can be covered in a short time, using interview worksheets and questionnaires. Often, the researcher wishes to obtain information about just a single variable, in which case a restricted set of questions may be used: a particular feature of pronunciation, for example, can be elicited by asking the informant to say a restricted set of words. There are also several direct methods of elicitation, such as asking informants to fill in the blanks in a substitution frame (e.g. I see a car), or feeding them the wrong stimulus for correction ('Is it possible to say / no can see?').
- G. A representative sample of language, compiled for the purpose of linguistic analysis, is known as a corpus. A corpus enables the linguist to make unbiased statements about frequency of usage, and it provides accessible data for the use of different researchers. Its range and size are variable. Some corpora attempt to cover the language as a whole, taking extracts from many kinds of text; others are extremely selective, providing a collection of material that deals only with a particular linguistic feature. The size of the corpus depends on practical factors, such as the time available to collect, process and store the data: it can take up to several hours to provide an accurate transcription of a few minutes of speech. Sometimes a small sample of data will be enough to decide a linguistic hypothesis; by contrast, corpora in major research projects can total millions of words. An important principle is that all corpora, whatever their size, are inevitably limited in their coverage, and always need to be supplemented by data derived from the intuitions of native speakers of the language, through either introspection or experimentation.

Questions 1-5

Complete the table below. Choose your **NO MORE THAN THREE WORDS** from the passage for each answer.

METHODS OF OBTAINING LINGUISTIC DATA	ADVANTAGES	DISADVANTAGES
(1)..... as informant	convenient	Method of enquiry not objective enough
Non-linguist as informant	Necessary with (2)..... and child speech	The number of factors to be considered
Recording an informant	Allows linguist's claims to be checked	(3)..... of sound
Videoing an informant	Allows speakers (4)..... to be observed	(5)..... might miss certain things

Practise Reading Passage 1.9

TELEPATHY

Since the 1970s, parapsychologists at leading universities and research institutes around the world have risked the derision of sceptical colleagues by putting the various claims for telepathy to the test in dozens of rigorous scientific studies. The results and their implications are dividing even the researchers who uncovered them.

Some researchers say the results constitute compelling evidence that telepathy is genuine. Other parapsychologists believe the field is on the brink of collapse, having tried to produce definitive scientific proof and failed. Sceptics and advocates alike do concur on one issue, however: that the most impressive evidence so far has come from the so-called ‘ganzfeld’ experiments, a German term that means ‘whole field’. Reports of telepathic experiences had by people during meditation led parapsychologists to suspect that telepathy might involve ‘signals’ passing between people that were so faint that they were usually swamped by normal brain activity. In this case, such signals might be more easily detected by those experiencing meditation-like tranquility in a relaxing ‘whole field’ of light, sound and warmth.

The ganzfeld experiment tries to recreate these conditions with participants sitting in soft reclining chairs in a sealed room, listening to relaxing sounds while their eyes are covered with special filters letting in only soft pink light. In early ganzfeld experiments, the telepathy test involved identification of a picture chosen from a random selection of four taken from a large image bank. The idea was that a person acting as a ‘sender’ would attempt to beam the image over to the ‘receiver’ relaxing in the sealed room. Once the session was over, this person was asked to identify which of the four images had been used. Random guessing would give a hit-rate of 25 per cent; if telepathy is real, however, the hit-rate would be higher. In 1982, the results from the first ganzfeld studies were analysed by one of its pioneers, the American parapsychologist Charles Honorton. They pointed to typical hit-rates of better than 30 per cent – a small effect, but one which statistical tests suggested could not be put down to chance.

The implication was that the ganzfeld method had revealed real evidence for telepathy. But there was a crucial flaw in this argument – one routinely overlooked in more conventional areas of science. Just because chance had been ruled out as an explanation did not prove telepathy must exist; there were many other ways of getting positive results. These ranged from ‘sensory leakage’ – where clues about the pictures accidentally reach the receiver – to outright fraud. In response, the researchers issued a review of all the ganzfeld studies done up to 1985 to show that 80 per cent had found statistically significant evidence. However, they also agreed that there were still too many problems in the experiments which could lead to positive results, and they drew up a list demanding new standards for future research.

After this, many researchers switched to autoganzfeld tests – an automated variant of the technique which used computers to perform many of the key tasks such as the random selection of images. By minimising human involvement, the idea was to minimise the risk of flawed

results. In 1987, results from hundreds of autoganzfeld tests were studied by Honorton in a ‘meta-analysis’, a statistical technique for finding the overall results from a set of studies. Though less compelling than before, the outcome was still impressive.

Yet some parapsychologists remain disturbed by the lack of consistency between individual ganzfeld studies. Defenders of telepathy point out that demanding impressive evidence from every study ignores one basic statistical fact: it takes large samples to detect small effects. If, as current results suggest, telepathy produces hit-rates only marginally above the 25 per cent expected by chance, it’s unlikely to be detected by a typical ganzfeld study involving around 40 people: the group is just not big enough. Only when many studies are combined in a meta-analysis will the faint signal of telepathy really become apparent. And that is what researchers do seem to be finding.

What they are certainly not finding, however, is any change in attitude of mainstream scientists: most still totally reject the very idea of telepathy. The problem stems at least in part from the lack of any plausible mechanism for telepathy.

Various theories have been put forward, many focusing on esoteric ideas from theoretical physics. They include ‘quantum entanglement’, in which events affecting one group of atoms instantly affect another group, no matter how far apart they may be. While physicists have demonstrated entanglement with specially prepared atoms, no-one knows if it also exists between atoms making up human minds. Answering such questions would transform parapsychology. This has prompted some researchers to argue that the future lies not in collecting more evidence for telepathy, but in probing possible mechanisms. Some work has begun already, with researchers trying to identify people who are particularly successful in autoganzfeld trials. Early results show that creative and artistic people do much better than average: in one study at the University of Edinburgh, musicians achieved a hit-rate of 56 per cent. Perhaps more tests like these will eventually give the researchers the evidence they are seeking and strengthen the case for the existence of telepathy.

Questions 1-10

Complete the table below. Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

Telepathy Experiments			
Name/ Date	Description	Result	Flaw
Ganzfeld studies 1982	Involved a person acting as a (1)..... who picked out one (2)..... from a random selection of four and a (3)..... who then tried to identify it.	Hit rates were higher than with random guessing	Positive results could be produced by factors such as (4)..... or (5).....
Autoganzfeld studies 1987	(6)..... were used for key tasks to limit the amount of (7)..... in carrying out the tests.	The results were then subjected to a (8).....	The (9)..... between different test results was put down to the fact that sample groups were not (10)..... (as with most ganzfeld studies).

Practice Reading Passage 1.10

ISSUES AFFECTING THE SOUTHERN RESIDENT ORCAS

- A. Orcas, also known as killer whales, are opportunistic feeders, which means they will take a variety of different prey species. J, K, and L pods (specific groups of orcas found in the region) are almost exclusively fish eaters. Some studies show that up to 90 percent of their diet is salmon, with Chinook salmon being far and away their favorite. During the last 50 years, hundreds of wild runs of salmon have become extinct due to habitat loss and overfishing of wild stocks. Many of the extinct salmon stocks are the winter runs of chinook and coho. Although the surviving stocks have probably been sufficient to sustain the resident pods, many of the runs that have been lost were undoubtedly traditional resources favored by the resident orcas. This may be affecting the whales' nutrition in the winter and may require them to change their patterns of movement in order to search for food.
- B. Other studies with tagged whales have shown that they regularly dive up to 800 feet in this area. Researchers tend to think that during these deep dives the whales may be feeding on bottomfish. Bottomfish species in this area would include halibut, rockfish, lingcod, and greenling. Scientists estimate that today's lingcod population in northern Puget Sound and the Strait of Georgia is only 2 percent of what it was in 1950. The average size of rockfish in the recreational catch has also declined by several inches since the 1970s, which is indicative of overfishing. In some locations, certain rockfish species have disappeared entirely. So even if bottomfish are not a major food resource for the whales, the present low numbers of available fish increases the pressure on orcas and all marine animals to find food. (For more information on bottomfish see the San Juan County Bottomfish Recovery Program.)
- C. Toxic substances accumulate in higher concentrations as they move up the food chain. Because orcas are the top predator in the ocean and are at the top of several different food chains in the environment, they tend to be more affected by pollutants than other sea creatures. Examinations of stranded killer whales have shown some extremely high levels of lead, mercury, and polychlorinated hydrocarbons. Abandoned marine toxic waste dumps and present levels of industrial and human refuse pollution of the inland waters probably presents the most serious threat to the continued existence of this orca population. Unfortunately, the total remedy to this huge problem would be broad societal changes on many fronts. But because of the fact that orcas are so popular, they may be the best species to use as a focal point in bringing about the many changes that need to be made in order to protect the marine environment as a whole from further toxic poisoning.
- D. The waters around the San Juan Islands are extremely busy due to international commercial shipping, fishing, whale watching, and pleasure boating. On a busy weekend day in the summer, it is not uncommon to see numerous boats in the vicinity of the whales as they travel through the area. The potential impacts from all this vessel traffic with regard to the whales and other marine animals in the area could be tremendous.

- E. The surfacing and breathing space of marine birds and mammals is a critical aspect of their habitat, which the animals must consciously deal with on a moment-to-moment basis throughout their lifetimes. With all the boating activity in the vicinity, there are three ways in which surface impacts are most likely to affect marine animals : (a) collision, (b) collision avoidance, and (c) exhaust emissions in breathing pockets.
- F. The first two impacts are very obvious and don't just apply to vessels with motors. Kayakers even present a problem here because they're so quiet. Marine animals, busy hunting and feeding under the surface of the water, may not be aware that there is a kayak above them and actually hit the bottom of it as they surface to breathe.
- G. The third impact is one most people don't even think of. When there are numerous boats in the area, especially idling boats, there are a lot of exhaust fumes being spewed out on the surface of the water. When the whale comes up to take a nice big breath of "fresh" air, it instead gets a nice big breath of exhaust fumes. It's hard to say how greatly this affects the animals, but think how breathing polluted air affects us (i.e., smog in large cities like Los Angeles, breathing the foul air while sitting in traffic jams, etc).
- H. Similar to surface impacts, a primary source of acoustic pollution for this population of orcas would also be derived from the cumulative underwater noise of vessel traffic. For cetaceans, the underwater sound environment is perhaps the most critical component of their sensory and behavioral lives. Orcas communicate with each other over short and long distances with a variety of clicks, chirps, squeaks, and whistles, along with using echolocation to locate prey and to navigate. They may also rely on passive listening as a primary sensory source. The long-term impacts from noise pollution would not likely show up as noticeable behavioral changes in habitat use, but rather as sensory damage or gradual reduction in population health. A new study at The Whale Museum called the Sea Sound Remote Sensing Network has begun studying underwater acoustics and its relationship to orca communication.

Questions 1-8

Complete the table chart below. Choose **NO MORE THAN THREE WORDS** for each answer.

Cause	Effect
Scientists believe some whales feed (1)..... .	These whales dive very deep.
Scientists believe that the area is being over fished.	Rockfish caught today is (2)..... than rockfish caught in the past.
Orcas are at the top of the ocean food chain.	(3)..... affects orcas more than it does other sea animals.
Orcas are a (4) species.	We can use orcas to make society aware of the problem of marine pollution.
Poopio enjoy boating, fishing, and whale watching in tho San Juan Islands.	On weekends there are (5)..... near the whales
Kayaks are (6).....	Marine animals hit them when they come up for air.
A lot of boats keep their motors running.	Whales breathe (7).....
Boats are noisy.	Whales have difficulty (8).....

ANSWER KEY

PASSAGE 1.1 1. TASTE BUDS 2. BALEEN / THE BALEEN WHALES 3. FORWARD, DOWNWARD (CAN BE WRITTEN IN ANY ORDER) 4. FRESHWATER DOLPHINS 5. WATER 6. LOWER FREQUENCIES 7. BOWHEAD, HUMPBACK	PASSAGE 1.2 1. SUDAN, INDIA (CAN BE WRITTEN IN ANY ORDER) 2. BICYCLES 3. SHOE SHINE COLLECTIVE 4. LIFE SKILLS	PASSAGE 1.3 1. SELF HELP 2. RELAXATION TRAINING 3. SLEEP QUALITY 4. BENZODIAZEPINES
	PASSAGE 1.4 1. TEMPERATE 2. EARLY SPRING 3. 2-5/ TWO-FIVE 4. SUB-TROPICAL 5. SOUTH AFRICAN TUNNELLING	PASSAGE 1.5 1. SOCIAL STATUS 2. LOCATION 3. UNIFORM 4. ORGANISATIONS
PASSAGE 1.6 1. 1772 2. 1781 3. 1787 4. 1977 5. 1986	PASSAGE 1.7 1. G 2. C 3. F 4. B	PASSAGE 1.8 1. THE LINGUIST ACT 2. FOREIGN LANGUAGES 3. THE QUALITY 4. NON-VERBAL BEHAVIOUR 5. VIDEO CAMERA

PASSAGE 1.9 1. SENDER 2. IMAGE 3. RECEIVER 4. SENSORY LEAKAGE (CAN BE WRITTEN IN ANY ORDER) 5. OUTRIGHT FRAUD (CAN BE WRITTEN IN ANY ORDER) 6. COMPUTERS 7. HUMAN INVOLVEMENT 8. META-ANALYSIS 9. LACK OF CONSISTENCY 10. LARGE ENOUGH	PASSAGE 1.10 1. BOTTOMFISH 2. LESS 3. POLLUTION 4. POPULAR 5. NUMEROUS BOATS 6. QUIET 7. EXHAUST FUMES 8. COMMUNICATING	
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